AI-Driven Transformation in Banking GRC: Implications for Workforce Management in UK and EU Banks

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AI is entering bank risk and compliance faster than people can govern it

75%

of UK financial firms already use AI

46%

have only a partial understanding of the AI they use

57%

say talent and skills are a constraint

Adoption is not the problem. **Understanding and skills are.**

Source: Bank of England & FCA (2024), Artificial intelligence in UK financial services

This is a management problem, and regulation makes it urgent

Four decisions managers now face

Which GRC roles to **redesign, reduce or create**

Whether to **build, buy or borrow** skills

How to **shape the team**

How to get staff to **trust AI appropriately**

Regulation

DORA	board & staff ICT competence
EU AI Act	AI literacy, human oversight
PRA SS1/23	skilled model validators
SM&CR	named personal accountability

Competence and human oversight are now legal obligations

Objectives: build the evidence, then turn it into a framework

Primary: examine how AI in GRC is reshaping workforce decisions (role structure, skills, staffing, employee adaptation) and develop a practical framework for management.

i. Which GRC activities are automated or augmented, and the effect on role demand

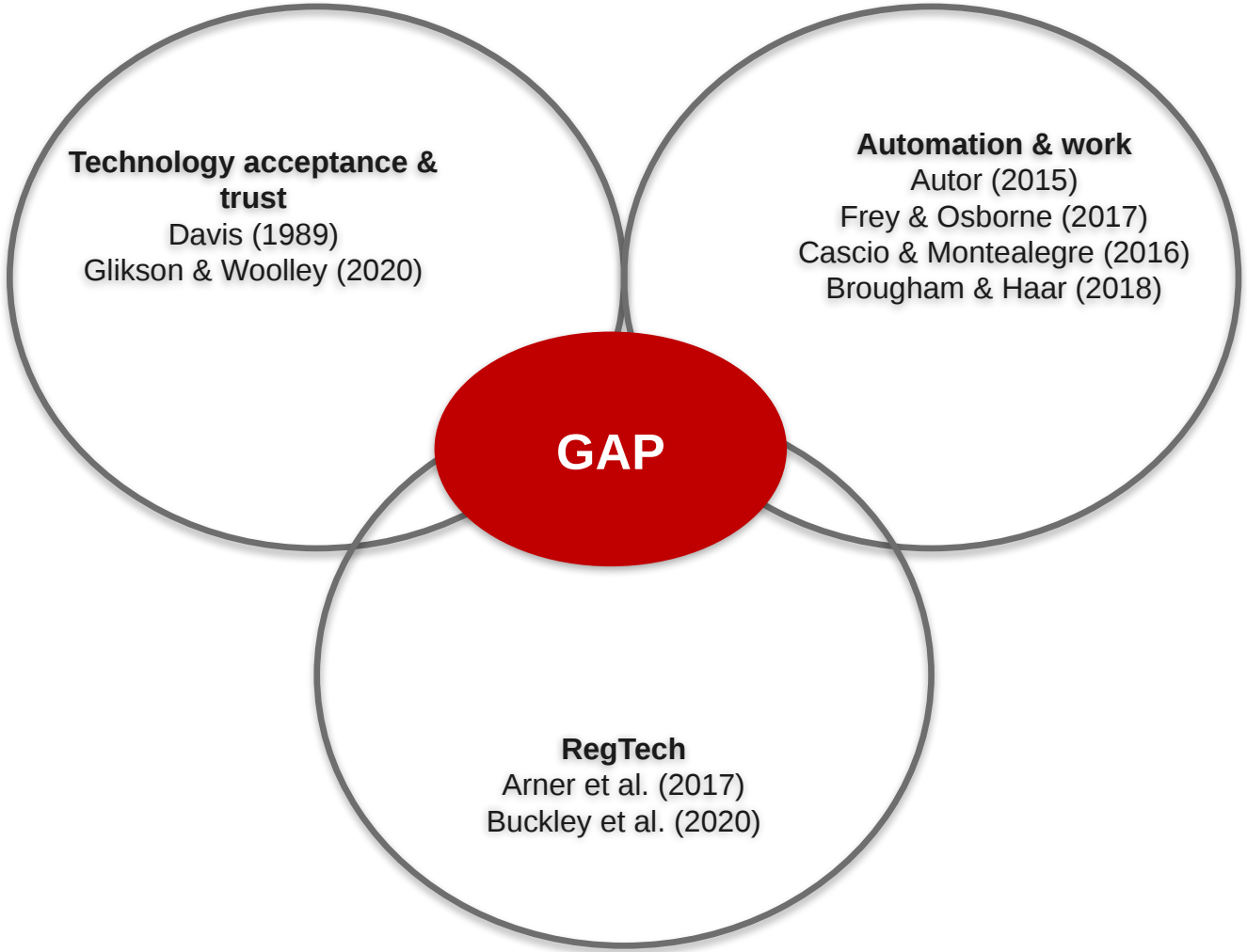
ii. How DORA, the AI Act, SS1/23 and SM&CR shape oversight and competence

iii. Emerging skill gaps and reskilling strategies

iv. Employee responses: usefulness, trust, replaceability

v. **An integrated workforce transformation framework and recommendations**

Three bodies of research, and none of them looks at the GRC workforce



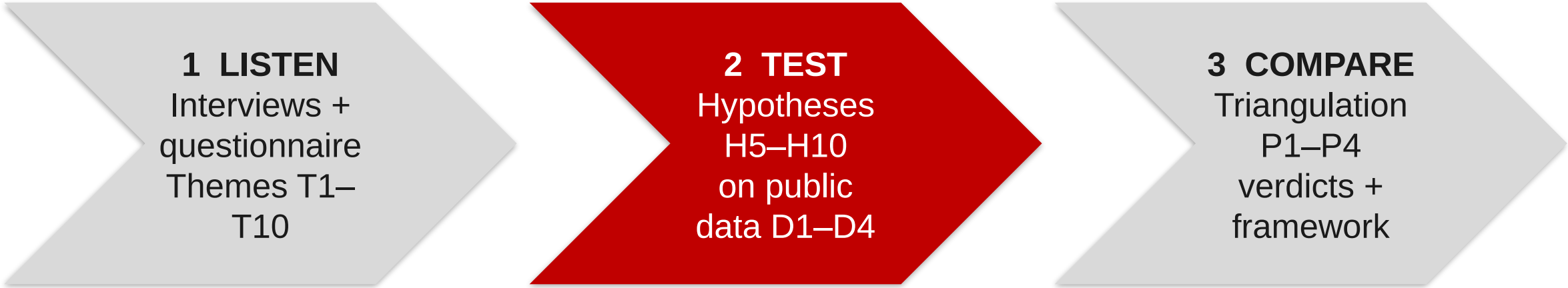
Research gap

Workforce management inside regulated banking GRC: roles, skills, staffing and adaptation

The literature gave four propositions to test

	Proposition	Grounded in
P1	Routine GRC tasks are automated first; judgement work is augmented	Autor (2015); Frey & Osborne (2017)
P2	Regulation turns competence and oversight into obligations	Buckley et al. (2020); DORA, AI Act, SS1/23, SM&CR
P3	AI needs a hybrid technical + regulatory skill profile	Cascio & Montealegre (2016)
P4	Losing expert knowledge is a hidden restructuring risk **EMERGENT**	Interviews (extends Autor, 2015)

A sequential exploratory mixed-methods design



The interviews decided **what** to test. The public data showed **whether it holds** across the industry.

The public datasets are independent sources; they were not produced from the interviews.

Two kinds of evidence: practitioners, and four public datasets

Primary

Written semi-structured interviews: 11 questions, 5 sections

Questionnaire: 25 Likert items, 7 constructs adapted from Davis; Glikson & Woolley; Brougham & Haar

Sampling: purposive + snowball; GRC professionals exposed to AI

Secondary (public)

D1 — **Eurostat**
AI adoption, 27 EU countries, 2021–2025

D2 — **Bank annual reports**
Large UK/EU banks, FY2022 vs FY2025; 19 paired

D3 — **Reed job adverts**
163 UK financial-services GRC adverts

D4 — **Published surveys**
BoE/FCA 2022 & 2024, IIA, Thomson Reuters

Every hypothesis traces back to something practitioners said

Interview theme	Questionnaire signal	Public-data hypothesis
T1 AI spreading, still pilot-heavy	Role will change: 4.33	H5a, H8: adoption is rising
T2/T3 Routine work absorbed	Usefulness: 3.58	H10a AI mentions; H6 skill mix in adverts
T4/T5 Humans must challenge AI	Can defend AI decision: 4.33	H10b: AI governance bodies
T6 Regulation felt unevenly	Oversight clear: 3.67	H10a regulation mentions; H10d UK vs EU
T7 Hybrid skills	Data skills more important: 4.00	H5b, H6, H7: skills and pay
T8/T9 Training lags, knowledge at risk	Training adequate: 3.33	H9 skills constraint; H10c reskilling vs AI
T10 Anxiety, weak change management	Replaceability concern: 3.89	Not testable on public data

Each test was chosen by the shape of the data

Same countries/banks measured twice	**Paired t-test + Wilcoxon; Friedman (4 yrs)**	H5a, H10a
Yes/no for the same banks, two years	**Exact McNemar**	H10b
Two independent, skewed groups	**Mann–Whitney U**	H7, H10d
Yes/no skill flags across two groups	**Chi-square / Fisher + Holm**	H6
Only published percentages	**Two-proportion & one-sample z**	H8, H9
Strength of a link	**Pearson / Spearman / OLS**	H5b, H10c

Questionnaire: descriptive only (means/SD vs neutral 3); significance tests are not meaningful on a small purposive sample

Practitioners: routine work goes; judgement and knowledge matter more

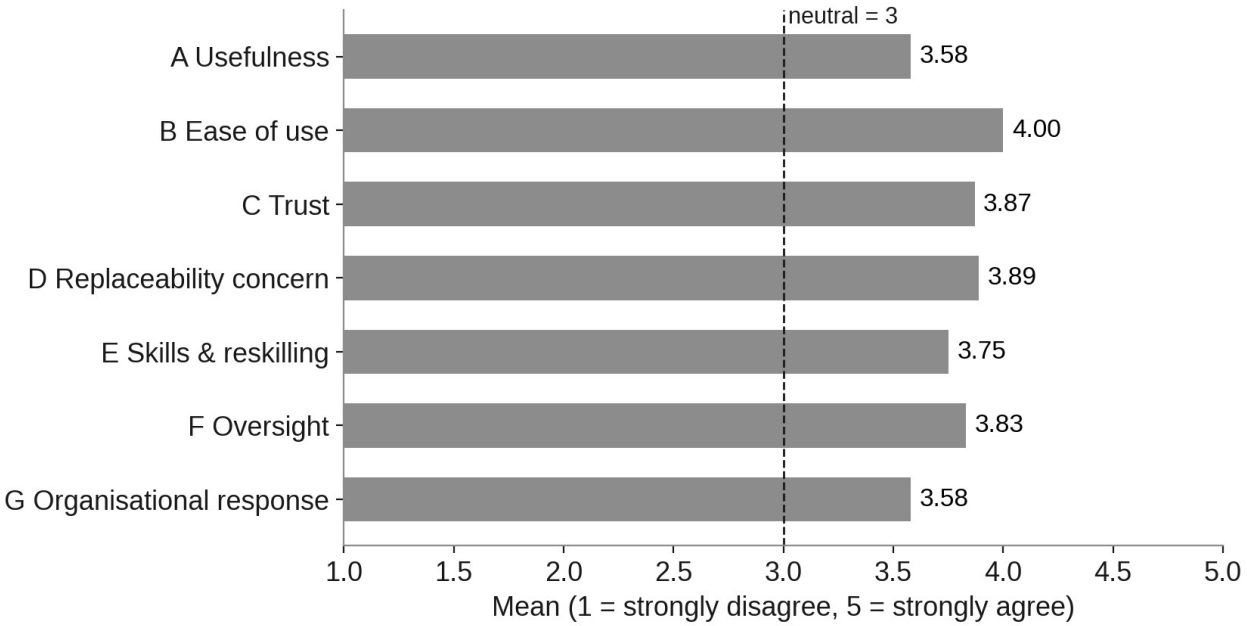
"GRC testing roles have disappeared."

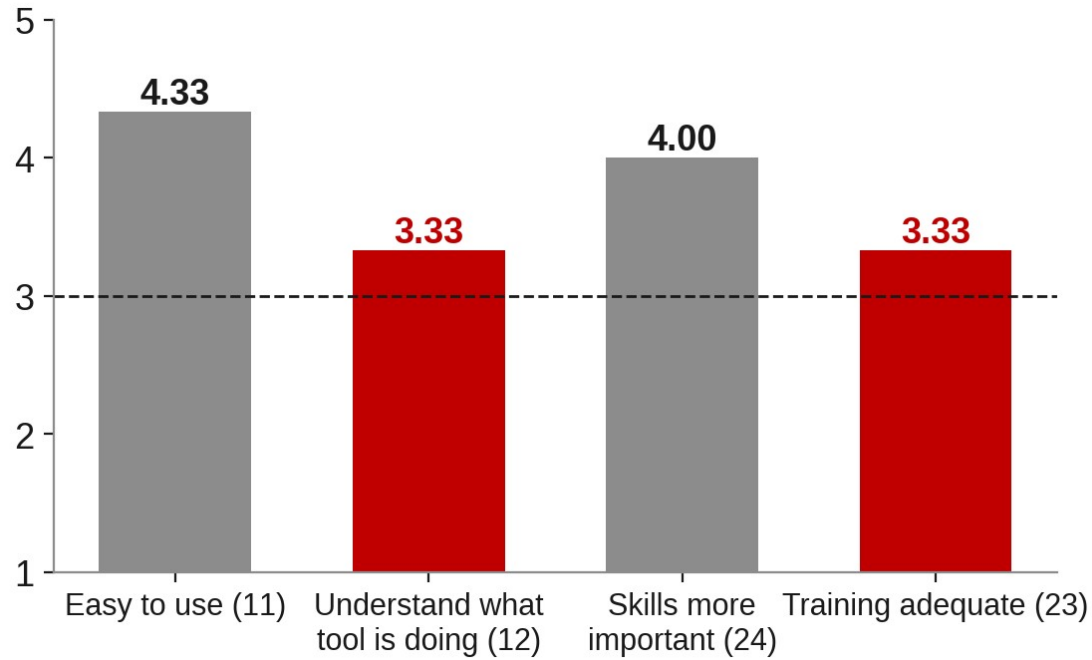
"C-level execs really do not understand what the AI is, and thus are likely to simply trust it."

"Too much business knowledge exists in people's heads… I would prioritise knowledge capture and transfer before any significant workforce reductions."

Ten themes by objective: (i) automation precedes AI · routine absorbed · roles recomposed (ii) expert review · accountability gap · regulation uneven (iii) hybrid skills · **knowledge loss** · build-buy-borrow (iv) ease welcomed, anxiety, weak change management

Questionnaire: easy to use but hard to see into, and training lags



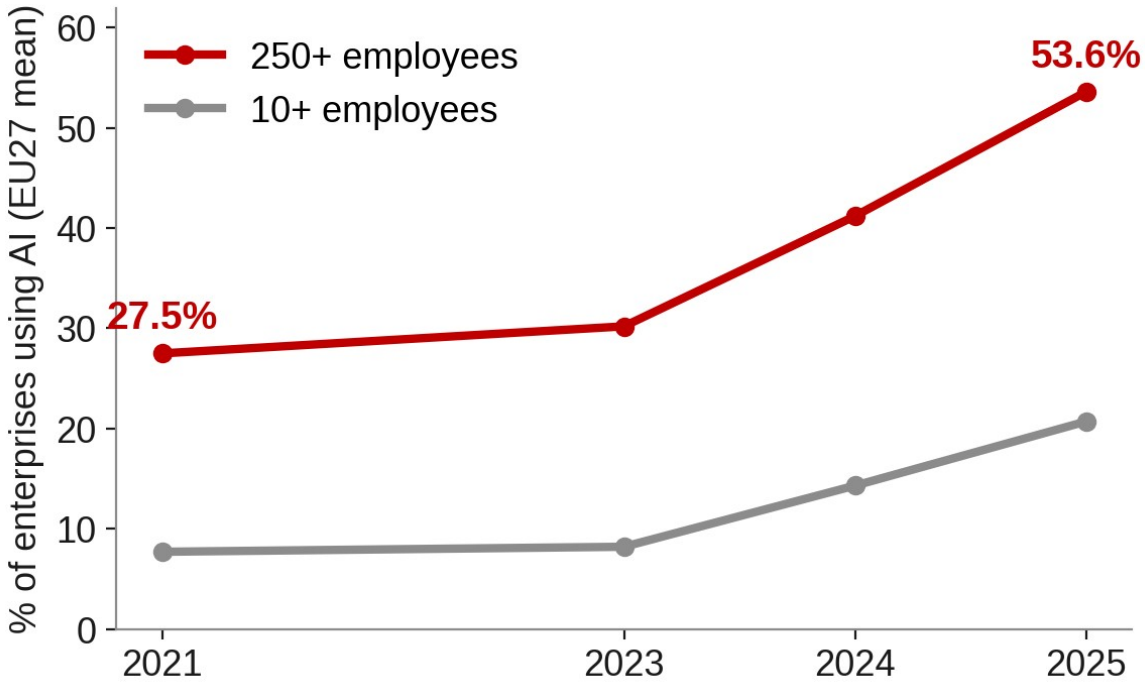


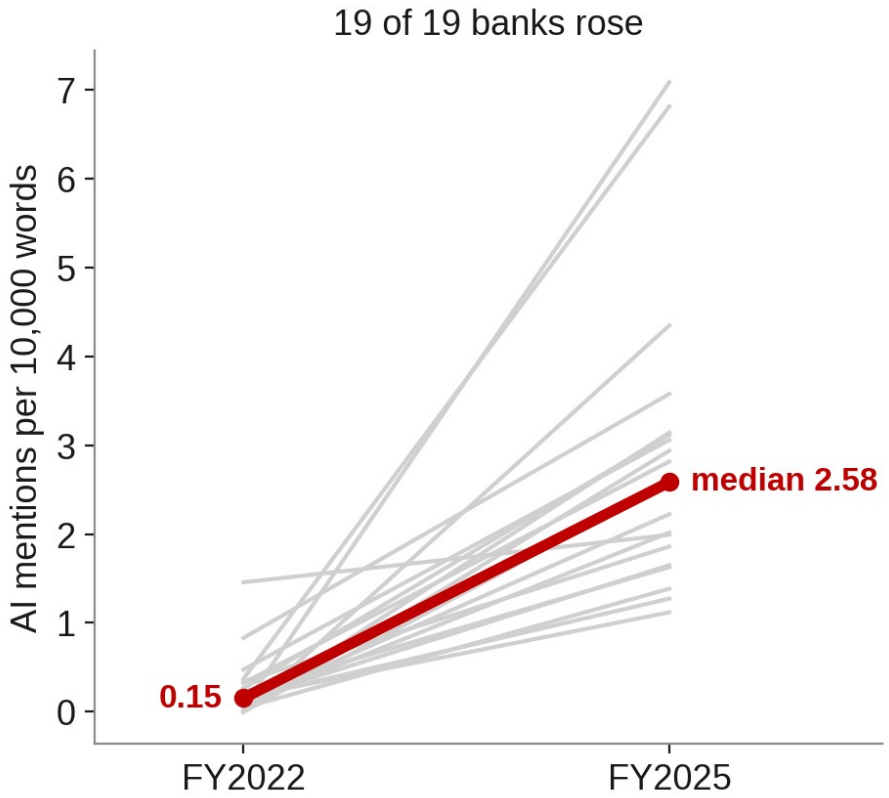
Usability–transparency gap: 4.33 vs 3.33

Training gap: skills matter more (4.00), training adequate 3.33

Descriptive means only; not generalised

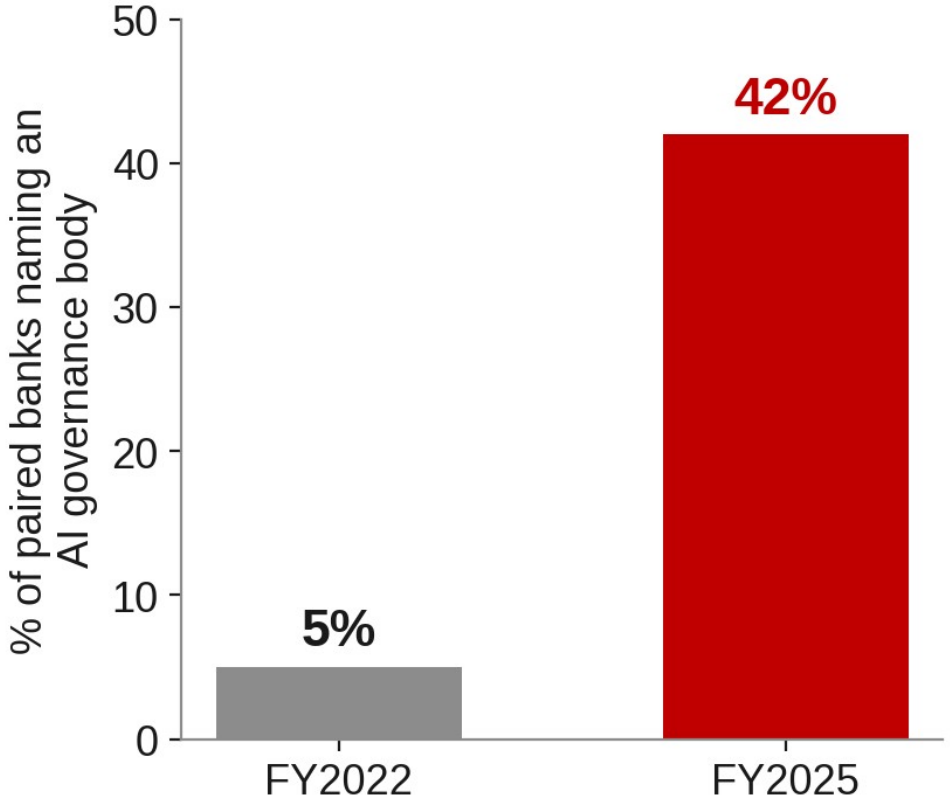
AI adoption is surging everywhere (H5a, H8, H10a supported)





EU large firms **27.5%** → **53.6%**, up in 27/27 countries (p < 0.0001) | UK FS firms **58%** → **75%** (p = 0.008) | Bank AI mentions **≈10×**, up in **19 of 19 banks** (p < 0.0001)

Governance structures are growing faster than governance practice



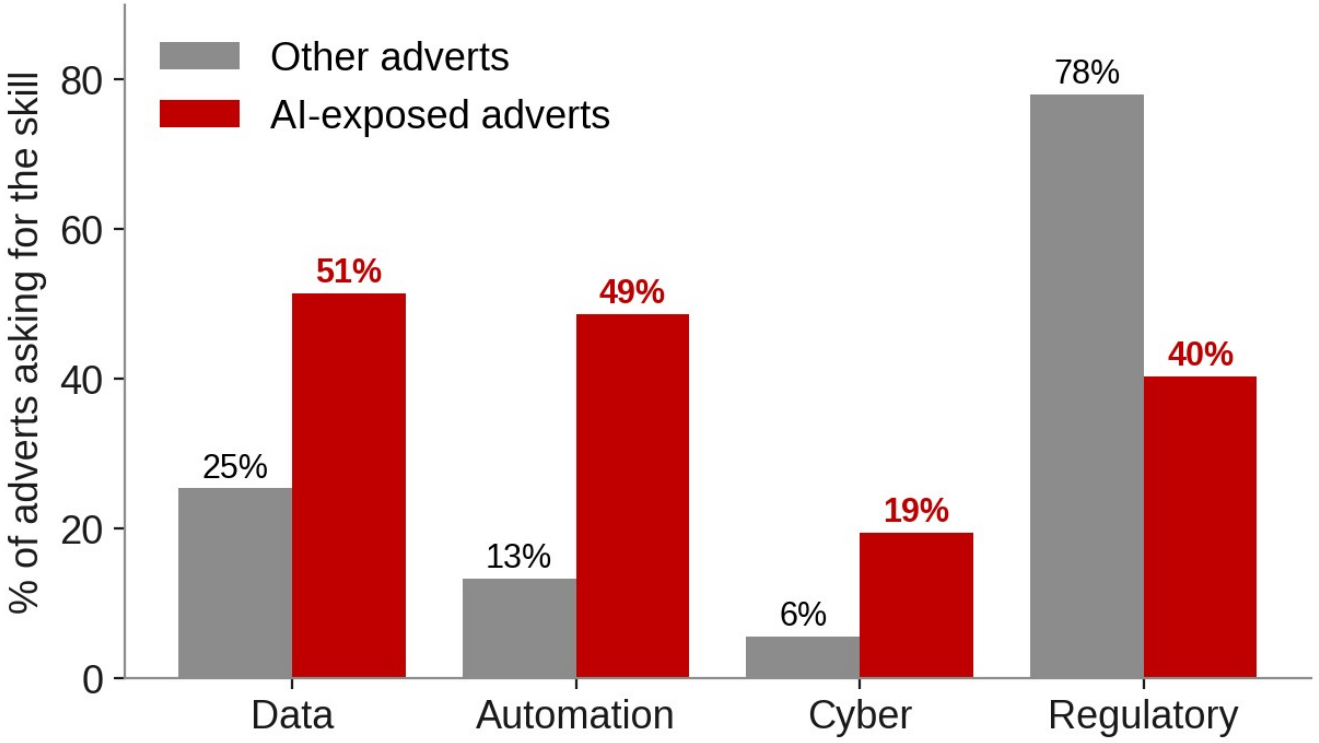
Banks naming an AI governance body: 5% → 42% (7 added, 0 dropped; exact McNemar p = 0.016)

AI Act / DORA / SS1/23 mentions: near zero → routine; 18 of 19 banks up ($p < 0.0001$)

Broader AI-governance wording flat ($p \approx 0.07$)

UK > EU on AI and AI-governance mentions (Holm p = 0.007 and 0.0004)

Skills are shifting, but reskilling is not keeping up



AI-exposed adverts want **more data, automation and cyber** skills but **fewer regulatory** skills (all Holm $p < 0.05$): the hybrid gap

AI-titled roles: median **£90k vs £60k** ($p = 0.009$; small group)

Reskilling mentions not linked to AI intensity ($\rho = -0.16$, $p = 0.49$)

Hypothesis scorecard

Hypothesis	Verdict
H1–H3 Useful, easy to use, roles will change (questionnaire)	● Supported (descriptive)
H4 Trust is conditional	◐ Partly
H5a, H5b EU adoption rising; ICT skills linked	● Supported
H6 Skill mix differs in AI adverts	● Supported
H7 AI roles pay more	◐ Partly (AI-titled only)
H8 UK FS adoption rose	● Supported
H9 Skills a majority constraint	◐ Partly
H10a AI, reskilling, regulation mentions rose	● Supported
H10a AI-governance wording rose	○ Not supported
H10b More AI governance bodies	● Supported
H10c AI and reskilling linked	○ Not supported
H10d UK and EU differ	◐ Partly (AI, AI governance)

Triangulation: all four propositions hold

	Interviews	Questionnaire	Public data	Verdict
P1 Routine automated first	T1–T3	Role change 4.33	H5a, H8, H10a	Supported
P2 Regulation → obligations	T4–T6	Items 26–27	H10a reg, H10b, H10d	Texts & structures; partly in practice
P3 Hybrid skills	T7, T9	Items 24–25	H6, H5b, H9	Supported
P4 Knowledge-loss risk	T8	–	Indirect (H10c)	Qualitative; emergent contribution

AI changes the composition of GRC work, not just its volume

AUTOMATED
Data consolidation · reconciliation · MI · first-level alert triage · periodic KYC refresh

AUGMENTED
Alert investigation · KYC/CDD judgement · model validation · audit fieldwork · regulatory change

NEWLY CREATED
AI governance lead · AI model-risk validator · AI audit lead · human-in-the-loop supervisor of AI agents

HUMAN-RETAINED
Interpreting regulatory intent · challenging decisions · assurance to boards · final accountability (SM&CR, DORA)

A seven-stage framework: protect knowledge before restructuring



Why this order? The experts whose routine work is automated are the same people needed to validate the AI.

Enablers: data quality & lineage · clear accountability map · workforce indicators tracked alongside AI indicators

Recommendations, conclusion and limits

GRC leaders

Map activities, not job titles

Capture knowledge before cutting roles

Make oversight operational

HR

Close the training gap: role-tiered AI literacy

Build, buy, borrow deliberately

Career paths for junior staff

Boards & senior managers

Close the AI-understanding gap at the top

Pair every AI plan with a people plan

Track workforce indicators

Conclusion: AI in GRC succeeds only if the people who answer for it can understand and challenge it.

Limits: small purposive primary sample (analytical generalisation) · disclosure ≠ action · 19 paired banks · one UK job board

Next: larger surveys · annual tracking of reports and adverts · EU job data · in-bank case studies

Thank you

Questions welcome

AI in GRC succeeds only if the people who answer for it can understand and challenge it.

Backup: D2 bank sample and coding

Both years (19): ABN AMRO, BBVA, Banco Santander, Bank of Ireland, Barclays, Danske Bank, Deutsche Bank, Erste Group, Groupe BPCE, HSBC, Intesa Sanpaolo, KBC, NatWest, Nationwide, Nordea, Santander UK, Société Générale, Standard Chartered, UniCredit

FY2025 only: Handelsbanken, ING, SEB | **FY2022 only:** Rabobank

Not coded: AIB, BNP Paribas, CaixaBank, Commerzbank, Crédit Agricole, Lloyds (no full English report / access blocked)

Coding: fixed keyword lists for AI, reskilling, AI governance and regulation; count ÷ total words × 10,000; binary flag for a named AI governance body. Each file checked against its title page.

Backup: full H10 results

Test	Statistic	p	Effect	n
AI density, paired t / Wilcoxon	t = 6.61 / W = 190	<0.0001	+2.61; 19/19 up	19
Reskilling density	t = 2.09 / W = 142	0.026 / 0.030	+0.16; 13/19 up	19
AI-governance density	t = 1.56 / W = 132	0.068 / 0.072	+0.15; 13/19 up	19
Regulation density	t = 5.42 / W = 171	<0.0001	+0.20; 18/19 up	19
AI governance body, exact McNemar	–	0.016	7 up, 0 down	19
AI vs reskilling, Spearman (FY2025)	ρ = −0.16	0.487		22
UK vs EU AI density, Mann–Whitney	U = 87	Holm 0.007	3.96 vs 2.00	6+16
UK vs EU reskilling	U = 23	Holm 0.140	0.22 vs 0.42	6+16
UK vs EU AI governance	U = 94	Holm 0.0004	2.01 vs 0.50	6+16
UK vs EU regulation	U = 45	Holm 0.858	0.21 vs 0.15	6+16

Backup: questionnaire item means (1–5)

Item	Mean
7 Improves performance	4.00
8 Increases volume	3.33
9 Improves quality	3.33
10 Useful overall	3.67
11 Easy to use	4.33
12 Understand what it does	3.33
13 Easy to learn	4.33
14 Understand outputs	3.67
15 Outputs reliable	3.67
16 Could defend decision	4.33
17 Rely without re-performing	3.67
18 Can override	4.00
19 Role could be done by AI	3.33

Item	Mean
20 Role will change	4.33
21 Career concern	4.00
22 Skills sufficient	3.67
23 Training adequate	3.33
24 Data skills more important	4.00
25 Judgement more important	4.00
26 Accountable people knowledgeable	4.00
27 Oversight clear in practice	3.67
28 Management communicated	4.00
29 Positive about AI	4.00
30 Would consider leaving	3.67
31 Committed to organisation	4.00

Backup: regulation mapped to workforce obligations

Instrument	Workforce requirement	Implication
DORA (2022/2554)	Board ICT knowledge (Art. 5); training for all staff (Art. 13)	Capability becomes auditable; training records = evidence
EU AI Act (2024/1689) + Omnibus	AI literacy (Art. 4); effective human oversight (Art. 14)	Role-based AI literacy; trained overseers
PRA SS1/23	Accountable SMF; documented lifecycle roles and validator skills	AI-capable model validation roles
SM&CR	Existing senior managers accountable for AI use	AI understanding gap = personal exposure
AMLR / AMLA (2024)	Compliance manager + AML compliance officer	Senior AML roles own AI-enabled monitoring

Backup: why the questionnaire is descriptive and public data carry the tests

Questionnaire: small purposive sample of senior practitioners, used to validate interview themes. Significance tests and Cronbach's alpha are not meaningful at this size, so means/SD are compared with the neutral point 3.

Interviews: free-text data, so thematic analysis is the right method.

Public data: large, independent samples (27 countries, 19 paired banks, 163 adverts, surveys of 71–6,506) allow proper inferential tests.

The interviews decided what to test; the public data tested whether it holds.